Applied Mathematics for Chemists
(화학수학)

Lecture 8

Multivariate Integral Calculus

중간고사

- 시간: 10월 20일(수) 오후 3시~4시
- 장소: 화학관 315호(≠ 교수계획표의 강의실)
- 계산기 불필요, 신분증 지참
- 10분 전까지 입실 완료

- 시험 범위: 다변수 미분(8장), 다변수 적분(9장)
- 수업 시간에 다루지 않은 내용은 범위에 포함되지 않음

Line Integral (선적분)

Consider a differential with two independent variables:

$$\mathrm{d}F = M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y$$

For the curve C joining (x_0, y_0) and (x_1, y_1),

the line integral (path integral) is defined as

$$\int_C \mathrm{d}F = \int_C [M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y]$$

Evaluation of Line Integral

$$\int_C \mathrm{d}F = \int_C [M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C \mathrm{d}F = \int_C [M\{x, y(x)\}\mathrm{d}x + N\{x(y), y\}\mathrm{d}y]$$

$$\int_C \mathrm{d}F = \int_{x_0}^{x_1} M\{x, y(x)\}\,\mathrm{d}x + \int_{y_0}^{y_1} N\{x(y), y\}\,\mathrm{d}y$$

Line Integrals of Exact Differentials

If dF is an exact differential, the line integral $\int_C dF$

- depends only on the initial and final points.
- equals to F(final point) $-$ F(initial point).

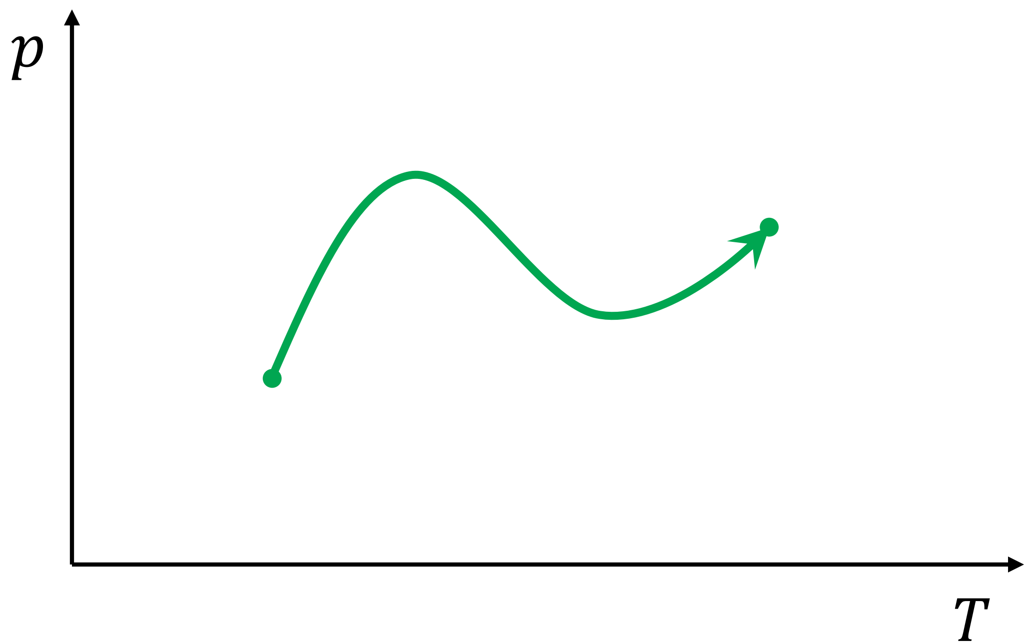
For an exact differential dF of the function $F(x, y)$,

with the initial point (x_0, y_0) and the final point (x_1, y_1),

$$\int_C \mathrm{d}F = \int_C \left[\left(\frac{\partial F}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial F}{\partial y}\right)_x \mathrm{d}y\right] = F(x_1, y_1) - F(x_0, y_0)$$

Line Integrals in Thermodynamics

- Equilibrium state: a point in a space
- Reversible process: a line integral



Double Integral (이중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f(x, y)\, \mathrm{d}y\, \mathrm{d}x$$

- Carry out the inner integration first
  - Treat the other independent variable (x) as a constant
- Then carry out the outer integration

Triple Integral (삼중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z)\, \mathrm{dz}\, \mathrm{d}y\, \mathrm{d}x$$

- First, integrate z from c_1 to c_2 (x, y = constant)
- Then integrate y from b_1 to b_2 (x = constant)
- Lastly, integrate x from a_1 to a_2

Not in Mortimer

Separation of Variables

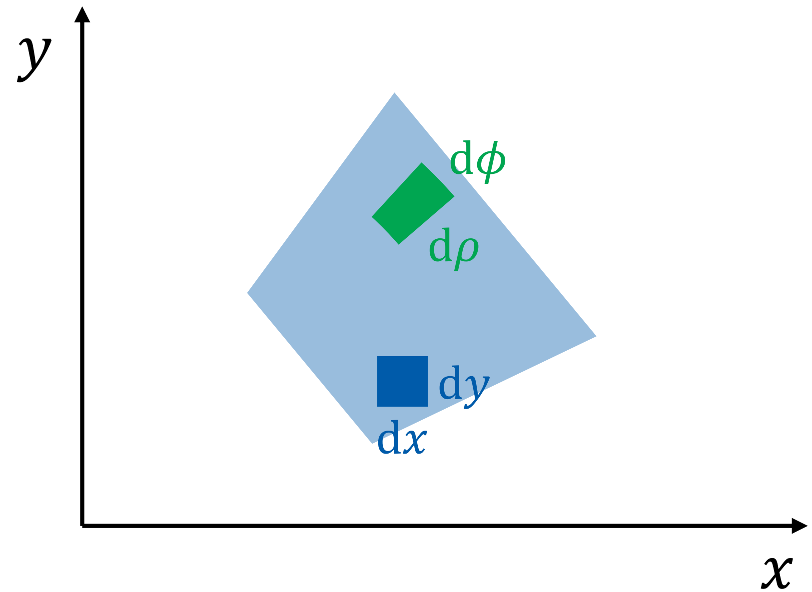
$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z)\, \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x$$

If $f(x, y, z) = f_x(x) f_y(y) f_z(z)$,

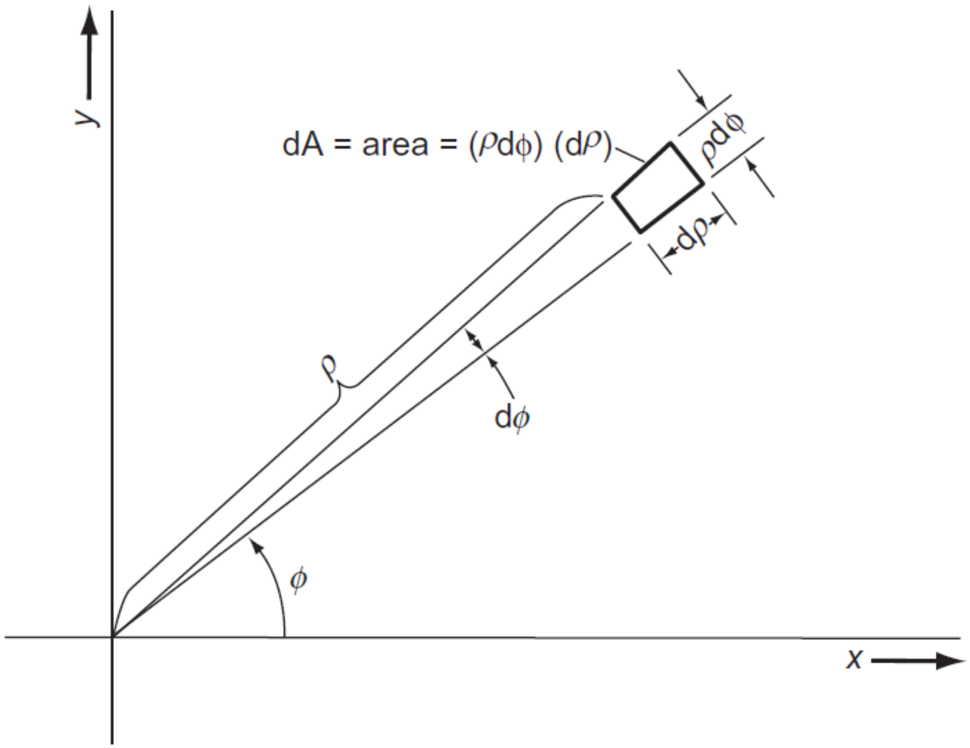
$$I = \left[\int_{a_1}^{a_2} f_x(x)\, \mathrm{d}x\right] \times \left[\int_{b_1}^{b_2} f_y(y)\, \mathrm{d}y\right] \times \left[\int_{c_1}^{c_2} f_z(z)\, \mathrm{d}z\right]$$

Changing Variables in Double Integrals

$$\int\int f(x,y)\,\mathrm{d}x\,\mathrm{d}y \underset{?}{\rightarrow} \int\int f(\rho,\phi)\,\mathrm{d}\rho\,\mathrm{d}\phi$$



Method 1: Geometric Method



$$\mathrm{d}A = \mathrm{d}x\mathrm{d}y = \rho\mathrm{d}\rho\mathrm{d}\phi$$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.4)

Conversion Formula

$$\int\int f\,\mathrm{d}A = \int\int f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int\int f(\rho,\phi)\rho\,\mathrm{d}\rho\,\mathrm{d}\phi$$

Today's Class

- Multiple integrals in different coordinate systems
  - Jacobian
  - In three dimensions

- Review: Multivariate differential calculus

Method 2: Jacobian

$$\int\int f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int\int f(\rho,\phi)\,\frac{\partial(x,y)}{\partial(\rho,\phi)}\,\mathrm{d}\rho\,\mathrm{d}\phi$$

where

$$\frac{\partial(x,y)}{\partial(\rho,\phi)} = \begin{vmatrix} \partial x/\partial\rho & \partial x/\partial\phi \\ \partial y/\partial\rho & \partial y/\partial\phi \end{vmatrix} = \left(\frac{\partial x}{\partial\rho}\right)\left(\frac{\partial y}{\partial\phi}\right) - \left(\frac{\partial x}{\partial\phi}\right)\left(\frac{\partial y}{\partial\rho}\right)$$

Jacobian Calculation

$$\frac{\partial(x,y)}{\partial(\rho,\phi)} = \left(\frac{\partial x}{\partial \rho}\right)\left(\frac{\partial y}{\partial \phi}\right) - \left(\frac{\partial x}{\partial \phi}\right)\left(\frac{\partial y}{\partial \rho}\right)$$

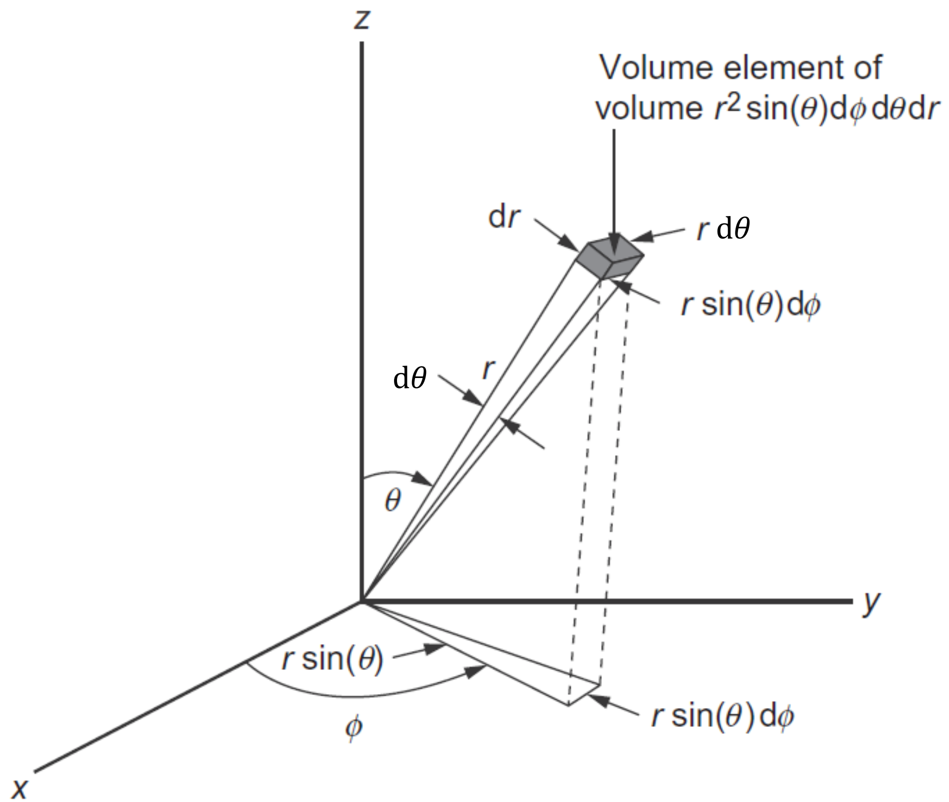
For $x = \rho \cos\phi$ and $y = \rho \sin\phi$,

$$\partial x/\partial\rho = \cos\phi\,,\ \ \partial x/\partial\phi = -\rho\sin\phi$$

$$\partial y/\partial\rho = \sin\phi\,,\ \ \partial y/\partial\phi = \rho\cos\phi$$

Jacobian = $(\cos\phi)(\rho\cos\phi) - (-\rho\sin\phi)(\sin\phi) = \rho$

Triple Integral in Spherical Coordinates



Jacobian: $r^2 \sin\theta$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.5)

Example 9.12

Evaluate the integral

$$I = \int_0^3 \int_0^\pi \int_0^{2\pi} r \cos^2 \phi \, r^2 \sin \theta \, \mathrm{d}\phi \, \mathrm{d}\theta \, \mathrm{d}r.$$

The integral can be decomposed:

$$I = \int_0^3 r^3 \, \mathrm{d}r \int_0^\pi \sin \theta \, \mathrm{d}\theta \int_0^{2\pi} \cos^2 \phi \, \mathrm{d}\phi$$

$$\int_0^3 r^3 \, \mathrm{d}r = \left[\frac{1}{4} r^4\right]_0^3 = \frac{81}{4}, \qquad \int_0^\pi \sin \theta \, \mathrm{d}\theta = [-\cos \theta]_0^\pi = 2$$

$$\int_0^{2\pi} \cos^2 \phi \, \mathrm{d}\phi = \int_0^{2\pi} \frac{\cos(2\phi) + 1}{2} \mathrm{d}\phi = \left[\frac{1}{4} \sin(2\phi) + \frac{1}{2} \phi\right]_0^{2\pi} = \pi$$

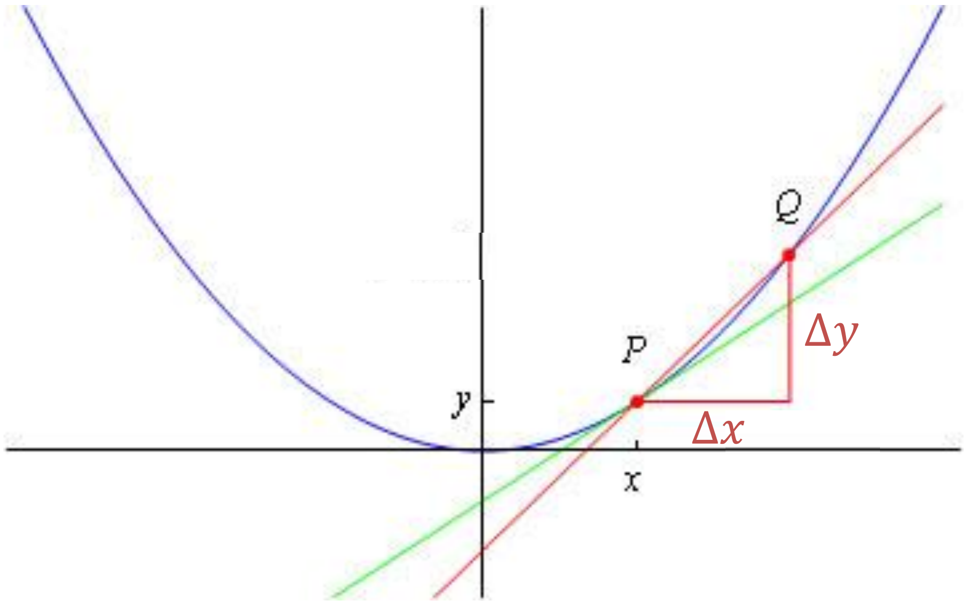
Hence, $I = 81\pi/2$.

Review:

Multivariate Differential Calculus

Differentiation

"The slope of a function"

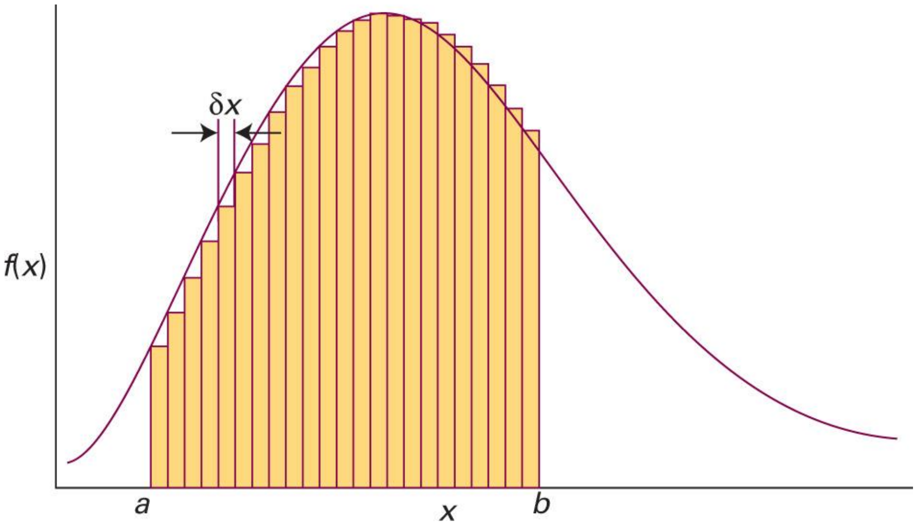


$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{\mathrm{d}y}{\mathrm{d}x}$$

(Image: http://wwwf.imperial.ac.uk/metric/metric_public/differentiation/index.html)

Integration

"The areas under curves"

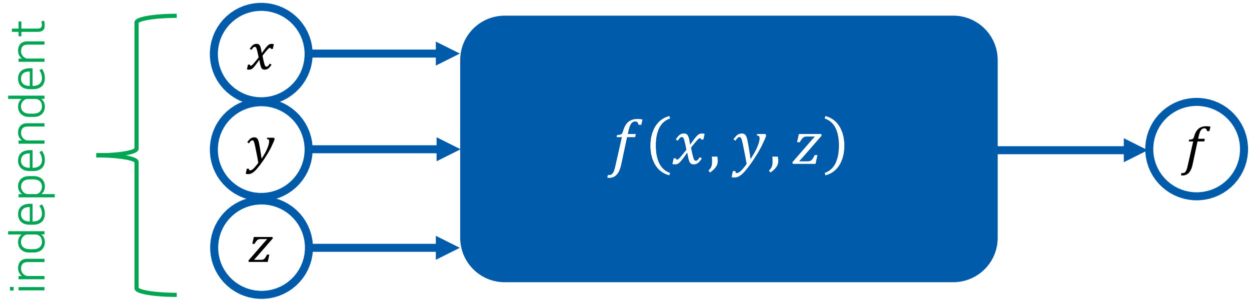


$$\lim_{\delta x \to 0} \sum_i f(x_i)\delta x = \int_a^b f(x)\,\mathrm{d}x$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 4, Sketch 1)

Function of Several Independent Variables

(다변수함수)



Example: $f(x, y, z) = x^2 + xy + 3z$

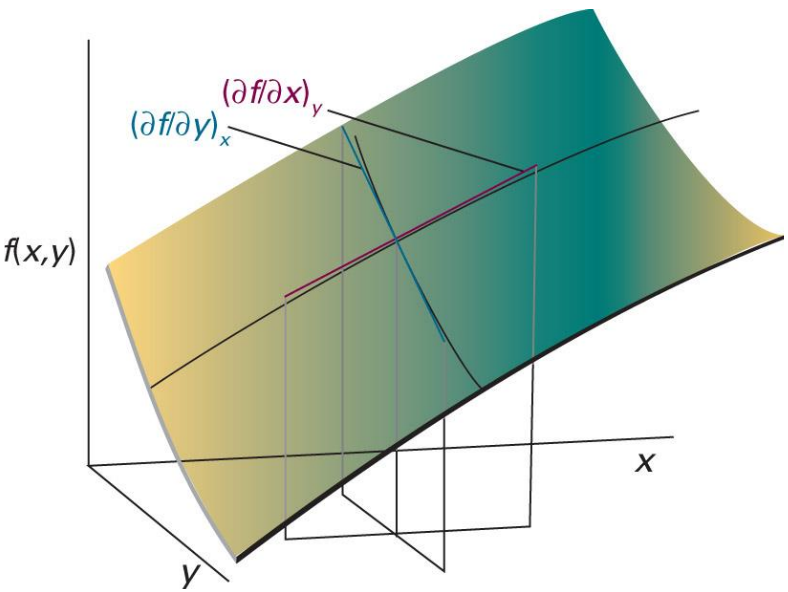
$$f(1,1,1) = 1 + 1 + 3 = 5$$

$$f(-1,1,0) = 1 - 1 + 0 = 0$$

$$\vdots$$

Partial Derivatives (편미분)

"the slope of the function with respect to one of the variables, all the other variables being held constant"



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 9, Sketch 1)

Fundamental Equation of Differential Calculus

(미분학의 기본 방정식)

- n independent variables: $x_1, x_2, \ldots, x_n$
- dependent variable: y

The differential of y is

$$\mathrm{d}y = \sum_{i=1}^{n} \left(\frac{\partial y}{\partial x_i}\right)_{x'} \mathrm{d}x_i$$

where x' means keeping all of the variables except for x_i fixed

Change of Variables

We can use different sets of independent variables:

- $U = U(T, V, n)$

$$\mathrm{d}U = \left(\frac{\partial U}{\partial T}\right)_{V,n} \mathrm{d}T + \left(\frac{\partial U}{\partial V}\right)_{T,n} \mathrm{d}V + \left(\frac{\partial U}{\partial n}\right)_{T,V} \mathrm{d}n$$

- $U = U(T, P, n)$

$$\mathrm{d}U = \left(\frac{\partial U}{\partial T}\right)_{P,n} \mathrm{d}T + \left(\frac{\partial U}{\partial P}\right)_{T,n} \mathrm{d}P + \left(\frac{\partial U}{\partial n}\right)_{T,P} \mathrm{d}n$$

Be Careful

For most systems,

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} \neq \left(\frac{\partial U}{\partial T}\right)_{P,n}$$

Fixed variables (subscripts) are important!

Variable-Change Identity

(변수-변화 항등식)

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \left(\frac{\partial P}{\partial T}\right)_{V,n}$$

Reciprocal Identity

(역 항등식)

$$\left(\frac{\partial y}{\partial x}\right)_{z,u} = \frac{1}{(\partial x/\partial y)_{z,u}}$$

Euler Reciprocity Relation

(오일러 역관계식)

For $z = z(x, y)$,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

Chain Rule

(연쇄 법칙)

For $z = z(u, x, y)$ and $x = x(u, v, y)$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Note:

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} \left(\frac{\partial y}{\partial z}\right)_{u,v} = 1$$

Check the subscripts!

Pfaffian Form: General Differential Form

(파피안형)

$$\mathrm{d}u = M(x, y)\mathrm{d}x + N(x, y)\mathrm{d}y$$

- Exact differential (완전 미분):

$$M(x, y) = \left(\frac{\partial u}{\partial x}\right)_y, \qquad N(x, y) = \left(\frac{\partial u}{\partial y}\right)_x$$

- Inexact differential (불완전 미분): otherwise

Test of Exactness

$$\mathrm{d}u = M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y$$

Using the Euler reciprocity relation,

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

if the differential is exact,

$$\left(\frac{\partial N}{\partial x}\right)_y = \left(\frac{\partial M}{\partial y}\right)_x$$

Integrating Factor
(적분인자)

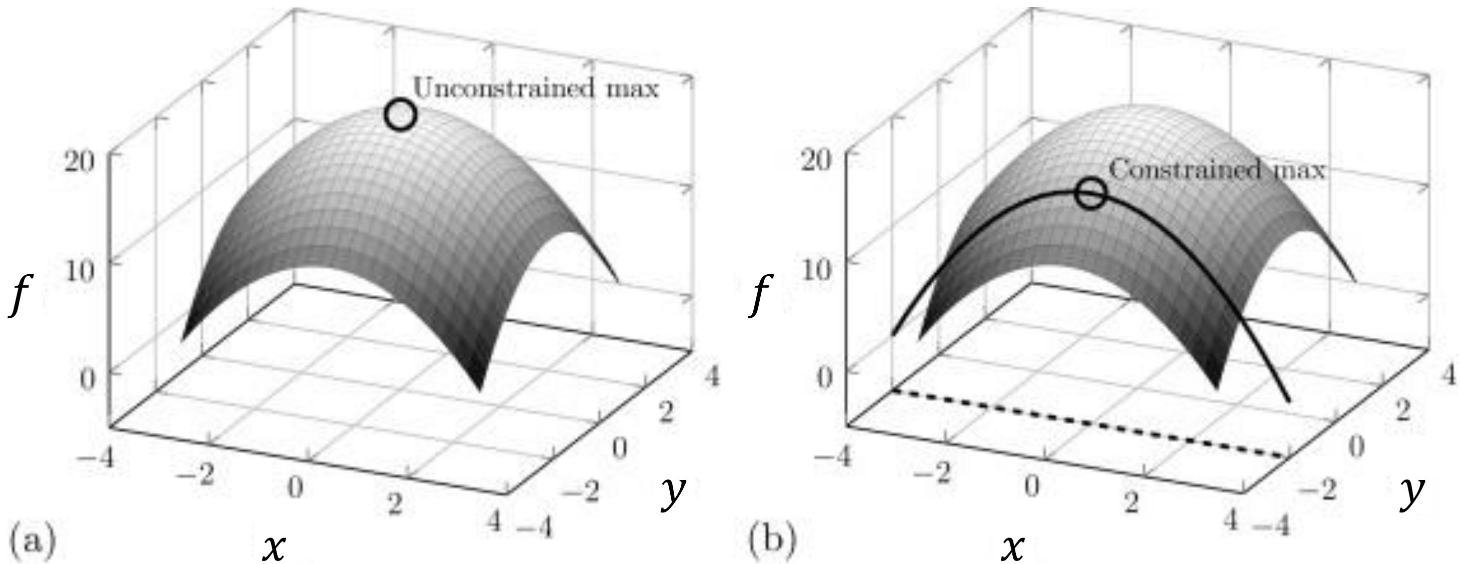
inexact differential → exact differential

by multiplying a function (integrating factor)

No general method to find an integrating factor

Constrained Optimization

(제약 최적화)



with a constraint (제약 조건)

(Image: Garrett, *Introduction to Actuarial and Financial Mathematical Methods* (2015), Figure 12.6)

Lagrange's Method

Find the constrained extremum in $f(x, y)$,

when the constraint is written in the form $g(x, y) = 0$

1. New function $u(x, y) = f(x, y) + \lambda g(x, y)$
2. Partial derivatives of u = 0
3. Solve the equations from point 2 and $g(x, y) = 0$ to obtain x, y, and λ → constrained extrema

Vector Derivatives

- Gradient (기울기) ∇f

  scalar → vector

- Divergence (발산) $\nabla \cdot \vec{F}$

  vector → scalar

- Curl (회전) $\nabla \times \vec{F}$

  vector → vector

- Laplacian (라플라스) $\nabla^2 f$

  scalar → scalar (2nd order)

Vector Derivatives

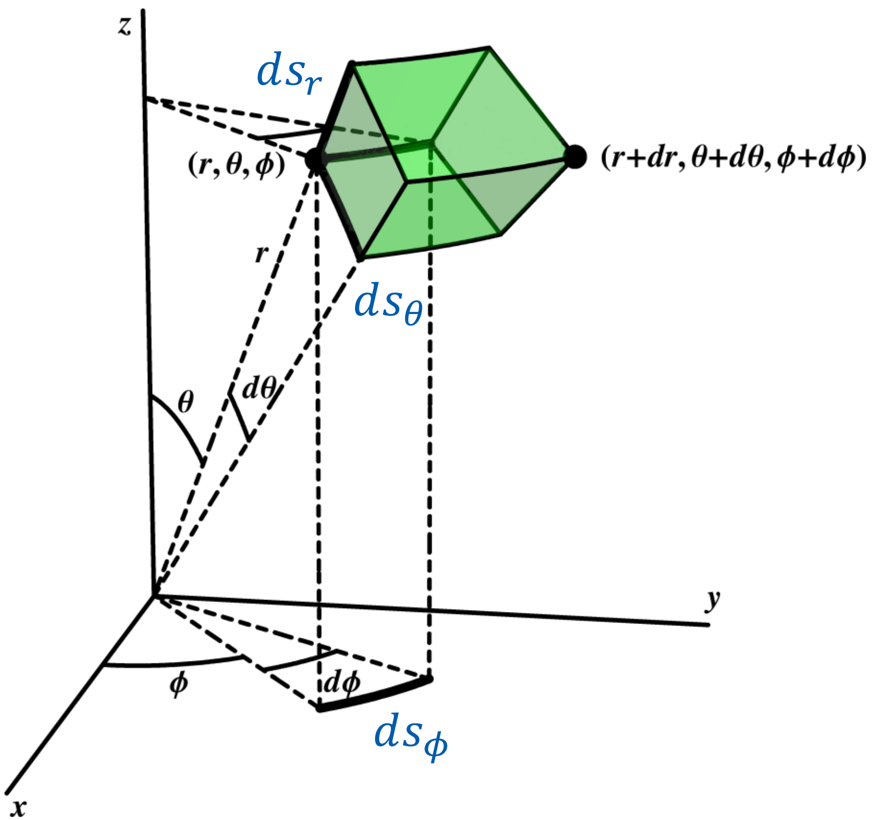
$$\nabla f = \hat{\imath}\left(\frac{\partial f}{\partial x}\right) + \hat{\jmath}\left(\frac{\partial f}{\partial y}\right) + \hat{k}\left(\frac{\partial f}{\partial z}\right)$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x}\right) + \left(\frac{\partial F_y}{\partial y}\right) + \left(\frac{\partial F_z}{\partial z}\right)$$

$$\nabla \times \vec{F} = \hat{\imath}\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \hat{\jmath}\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \hat{k}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2}\right) + \left(\frac{\partial^2 f}{\partial y^2}\right) + \left(\frac{\partial^2 f}{\partial z^2}\right)$$

Infinitesimal Displacements



$$ds_r = dr$$

$$ds_\theta = r d\theta$$

$$ds_\phi = r \sin\theta \, d\phi$$

(Image: https://bridge.math.oregonstate.edu/Book/dscoord.html)

Infinitesimal Displacements

In Cartesian coordinates,

$$d\vec{r} = \hat{\imath}dx + \hat{\jmath}dy + \hat{k}dz$$

In spherical polar coordinates,

$$d\vec{r} = \hat{e}_r dr + \hat{e}_\theta r d\theta + \hat{e}_\phi r \sin\theta \, d\phi$$

In general,

$$d\vec{r} = \hat{e}_1 h_1 dq_1 + \hat{e}_2 h_2 dq_2 + \hat{e}_3 h_3 dq_3$$

Vector Derivatives in Other Coordinates

$$\nabla f = \hat{e}_1 \frac{1}{h_1}\frac{\partial f}{\partial q_1} + \hat{e}_2 \frac{1}{h_2}\frac{\partial f}{\partial q_2} + \hat{e}_3 \frac{1}{h_3}\frac{\partial f}{\partial q_3}$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial(F_1 h_2 h_3)}{\partial q_1} + \frac{\partial(F_2 h_1 h_3)}{\partial q_2} + \frac{\partial(F_3 h_1 h_2)}{\partial q_3}\right]$$

$$\nabla \times \vec{F} = \hat{e}_1 \frac{1}{h_2 h_3}\left[\frac{\partial(h_3 F_3)}{\partial q_2} - \frac{\partial(h_2 F_2)}{\partial q_3}\right] + \hat{e}_2 \frac{1}{h_1 h_3}\left[\frac{\partial(h_1 F_1)}{\partial q_3} - \frac{\partial(h_3 F_3)}{\partial q_1}\right]$$

$$+ \hat{e}_3 \frac{1}{h_1 h_2}\left[\frac{\partial(h_2 F_2)}{\partial q_1} - \frac{\partial(h_1 F_1)}{\partial q_2}\right]$$

Vector Derivatives in Other Coordinates

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_1 h_3}{h_2}\frac{\partial f}{\partial q_2}\right) + \frac{\partial}{\partial q_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial q_3}\right)\right]$$